

INT531

Preparing the network

for the data-centre room and the containment rack

Supplement · pairs with weeks 2 to 5

School of Information Technology, King Mongkut's University of Technology Thonburi

Teach as one 3-hour block, or insert section by section | Content updated 2026



What this supplement covers

Ordered from hardest to change later, to easiest

01

Cable plant architecture

The standard layers, and where to put the access switch

02

Cable and connector types

Copper, fibre, DAC — matching distance to speed

03

Racks, containment and cabling

How airflow constrains the way you dress cables

04

ToR switches and uplinks

Selection, airflow direction, and oversubscription maths

05

VLANs and IP design

The room's plan, IPv4, IPv6 and a separate management network

06

Internet path and acceptance

Egress, redundancy, and measurable acceptance criteria

Learning outcomes for this supplement

By the end of this set you will be able to...

SLO N.1 Explain the standard cabling layers and choose a switch placement that suits the room

CLO4

SLO N.2 Select cable and connector types to match distance, speed and budget

CLO4

SLO N.3 Explain how airflow direction and cable dressing affect cooling performance

CLO4, CLO5

SLO N.4 Compute an uplink oversubscription ratio and judge whether it is acceptable

CLO5

SLO N.5 Design a VLAN plan and an addressing plan for both IPv4 and IPv6

CLO4

SLO N.6 Accept network work against measurable criteria, not just link lights

CLO4, CLO7



01

Cable plant architecture

Laid once and lived with for a decade — think it through before touching anything

The standard cabling layers

Per TIA-942 and ISO/IEC 11801-5 · the abbreviations differ, the idea does not

Layer	Full name	What it does	In our room
ER	Entrance Room	Where cabling from outside the building lands, and the demarcation with the provider	The building's comms room
MDA	Main Distribution Area	The room's main aggregation point, holding routers and core switches	The core rack in CB-201
HDA	Horizontal Distribution Area	Distribution to each row, where the patch panels live	The patch rack at the row end
EDA	Equipment Distribution Area	The racks holding actual servers, connected by short in-rack cables	Our containment rack
ZDA	Zone Distribution Area	An optional midpoint, used while the layout is still moving	Not used in this phase

The benefit of layering: moving a rack or adding servers touches only the EDA cabling, never the trunk that is already installed

Where does the switch go?

This one decision sets how much copper the whole room needs

Three ways to place the access switch

Top of Rack (ToR)

- สวิตช์อยู่ในตู้เดียวกับเซิร์ฟเวอร์
- สายทองแดงสั้นมาก อยู่ในตู้ทั้งหมด
- อับลิงก์ไฟเบอร์ออกจากตู้ 2-4 เส้น
- พอร์ตเหลือทิ้งถ้าตู้ไม่เต็ม

เหมาะกับห้องที่ตู้เต็มค่อนข้างสม่ำเสมอ

End of Row (EoR)

- สวิตช์รวมอยู่ตู้ปลายแถว
- สายทองแดงยาวข้ามตู้ ต้องเดินรางเยอะ
- ใช้พอร์ตคัมกว่า เพราะแชร์ทั้งแถว
- ตู้เสียหนึ่งตู้ไม่กระทบสวิตช์

เหมาะเมื่อตู้แต่ละใบมีเครื่องไม่มาก

Middle of Row (MoR)

- สวิตช์อยู่กลางแถว
- ระยะสายเฉลี่ยสั้นกว่า EoR
- ใช้พอร์ตคัมเหมือน EoR
- จัดวางสายซับซ้อนกว่า ToR

ทางสายกลางระหว่างสองแบบแรก

ห้องของเราเลือก ToR เพราะมี 20 กลุ่มที่ต้องแยกพอร์ตชัดเจน
และต้องการให้สายทองแดงทั้งหมดจบในตู้เดียวกัน เหลือแค่ไฟเบอร์ที่ออกจากตู้



02

Cable and connector types

Choose the wrong type and the distance will tell you on go-live day

Copper — choose by distance and speed

Distances are the standard's maximum channel length, patch cords at both ends included

Type	Speeds supported	Maximum length	When to use it
Cat5e	1 Gbps	100 m	Should not be installed new in a data centre
Cat6	1 Gbps at 100 m · 10 Gbps only to 37–55 m	100 m (at 1G)	General work with no 10 Gbps requirement
Cat6A	10 Gbps	100 m	The sensible default for any new installation
Cat8	25 and 40 Gbps	30 m	In-rack or rack-to-rack ToR links only

The Cat6 caveat: the advertised 10 Gbps is far short of 100 m and depends on alien crosstalk from neighbouring cables. For 10 Gbps at full length, use Cat6A

Fibre and connectors

Pick by distance first, then by budget

Type	Common jacket colour	Reach at 10G / 100G	Suited to
OM3 (multimode)	Aqua	300 m / 70 m (SR4)	Within one room, under about a hundred metres
OM4 (multimode)	Violet	400 m / 100 m (SR4)	The most common data-centre choice
OM5 (multimode)	Lime green	Similar to OM4, but supports multiple wavelengths	When SWDM is planned for later
OS2 (singlemode)	Yellow	10 km and beyond (LR)	Between buildings, or out to a provider
LC connector A small duplex connector used for 1G, 10G and 25G over one fibre pair — the most common one inside a rack			
MPO connector Bundles several fibres into one connector, used by 40G SR4 and 100G SR4 which need eight fibres at once			
Modules at both ends Every fibre run needs a transceiver at each end, and both must be the same type			

DAC, AOC or fibre with modules

Three ways to connect a server to the ToR, at very different prices

Option	Usable distance	Relative cost	The limitation to know
DAC (direct attach copper)	Up to 3–5 m passive	Cheapest	Thick and stiff, awkward to dress, limited bend radius
AOC (active optical cable)	About 3–30 m	Middle	Connectors are fixed to the cable; modules cannot be swapped
Fibre + transceivers	From a hundred metres to kilometres, by fibre type	Most expensive, most flexible	Two modules per run, and they must be compatible with the switch

In our containment rack: servers reach the ToR over DAC because they share the rack, while the ToR uplinks use fibre and modules because they leave the room



03

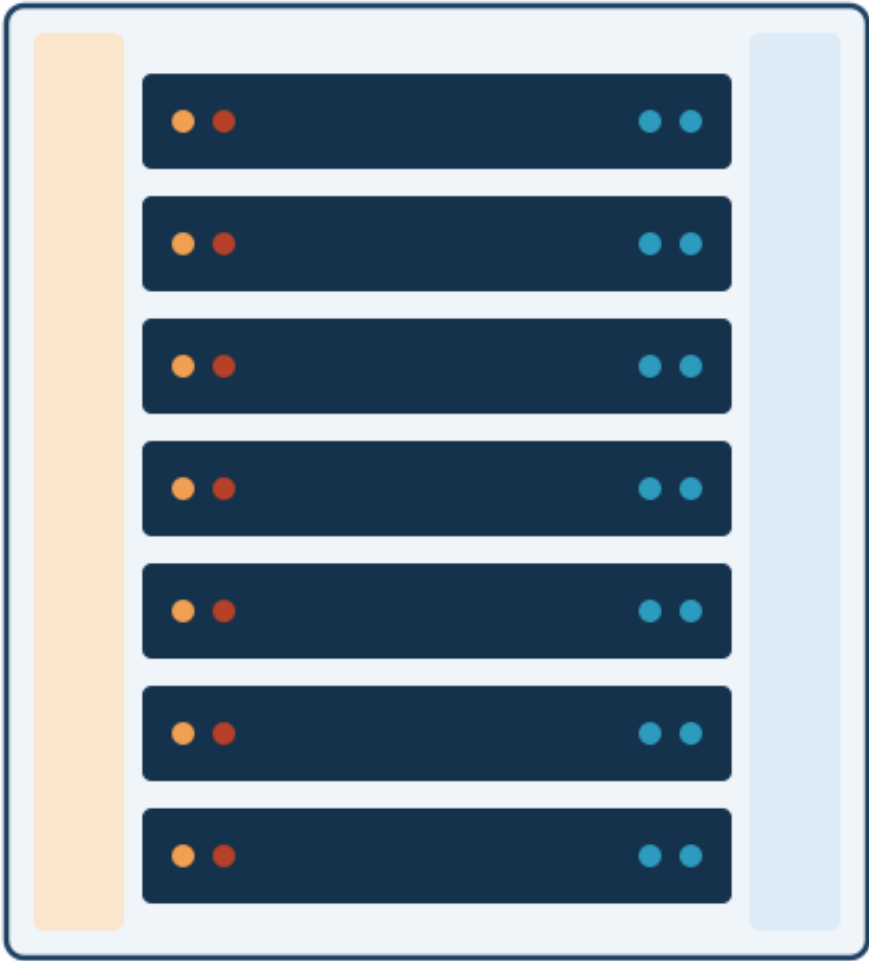
Racks, containment and cabling

Tidy cabling is not about tidiness — it is about airflow

Cable discipline behind the rack

The reasoning behind each rule

หลังตู้: สิ่งที่ทำให้ตู้อยู่ได้สิบปี หรือร้อยใหม่ในหกเดือน



สายไฟ (ซ้าย)

สายสัญญาณ (ขวา)

- แยกไฟกับสัญญาณ**
ลดสัญญาณรบกวนเหนี่ยวนำ และทำให้ถอดฝั่งใดฝั่งหนึ่งได้โดยไม่กระทบอีกฝั่ง
- เพื่อความยาวพอดี ไม่ยาวเกิน**
สายที่ยาวเกินต้องขดเก็บ ซึ่งกินพื้นที่และขวางทางลมออกด้านหลัง
- รักษารัศมีโค้งขั้นต่ำ**
ทองแดงประมาณ 4 เท่าของเส้นผ่านศูนย์กลาง ไฟเบอร์ประมาณ 10 เท่า
ถ้าหักเกินค่านี้ค่าสูญเสียจะเพิ่มขึ้นที่
- ติดป้ายทั้งสองปลายตาม TIA-606**
ป้ายปลายเดียวคือป้ายที่ใช้ไม่ได้
เพราะต้องรู้ทั้งต้นทางและปลายทางจากทั้งสองด้าน

ห้ามให้สายห้อยขวางด้านหลังตู้ เพราะเป็นการปิดทางลมร้อนออก
ซึ่งทำให้การกันลมที่ลงทุนไปทั้งหมดไม่ได้ผล

Airflow direction of network equipment

The most commonly missed detail when installing switches

Some switches draw air from the port side

Mount one with its ports facing the hot aisle and it inhales hot exhaust, then overheats within months

Airflow direction is a purchase decision

Vendors ship front-to-back and back-to-front variants under different part numbers; it must match the rack

Check before tightening the screws

Read the airflow label on the chassis and compare it with the rack's airflow before the unit is fixed

If a switch with the wrong airflow is unavoidable, use the vendor's airflow conversion kit — do not simply live with it



04

ToR switches and uplink design

Whether the ports are enough is a calculation, not a feeling

A checklist before choosing a ToR switch

Eight questions to answer before ordering

- | | | |
|---|------------------------------------|--|
| 1 | Access port count and speed | From the maximum servers per rack times ports per server, plus at least 20% headroom |
| 2 | Uplink count and speed | Decide the acceptable oversubscription ratio first, then work backwards |
| 3 | Airflow direction | Must match the rack's airflow, and must be stated on the order |
| 4 | PSU count | Two, on separate PDUs, as covered in week 3 |
| 5 | Chassis depth | Some switches are deep enough to foul the rear cable tray; check against the actual rack |
| 6 | A separate management port | Out of band from the data ports, so it is reachable when the main network is not |
| 7 | Full IPv6 support | Routing, DHCPv6 snooping and RA guard — not merely the ability to hold an IPv6 address |
| 8 | It must export metrics | SNMP or gNMI, so Prometheus can scrape it |

Computing oversubscription

Worked from our own rack

1

Sum the server-side bandwidth

20 servers × 2 ports × 25 Gbps

1,000 Gbps

2

Sum the uplink bandwidth

2 ToR × 2 uplinks × 100 Gbps

400 Gbps

3

Compute the ratio

1,000 ÷ 400

2.5 : 1

4

Check it against the target

General workloads tolerate about 3:1 · network-heavy work wants under 2:1

Acceptable for general use

The catch: lose one ToR and half the uplink capacity goes with it, making the ratio 5:1 — a number you must also be willing to accept



05

VLANs and IP address design

A good plan is one where reading the number tells you whose it is

The room's VLAN and addressing plan

VLAN numbers map directly onto group numbers, so they are easy to remember and audit

แผน VLAN และ IP ของห้อง — ออกแบบครั้งเดียว ใช้ทั้งปีการศึกษา

VLAN	ใช้ทำอะไร	IPv4	IPv6
101–120	กลุ่มนักศึกษา 1–20	10.20.G.0/24	2001:db8:a:1xx::/64
250	จัดการอุปกรณ์ (Management)	10.20.250.0/24	2001:db8:a:250::/64
251	จัดเก็บข้อมูล (Storage)	10.20.251.0/24	2001:db8:a:251::/64
254	นอกแถบสัญญาณ (OOB / IPMI)	10.20.254.0/24	fd00:20:254::/64
999	พอร์ตที่ยังไม่ใช้งาน (Blackhole)	ไม่กำหนด	ไม่กำหนด

พอร์ตสวิตช์ที่ยังไม่ใช้ต้องถูกโยนเข้า VLAN 999 และปิดไว้ · พอร์ตที่เปิดทิ้งไว้ใน VLAN ปกติ คือช่องทางเข้าเครือข่ายที่ไม่มีใครเฝ้า

Designing the IPv4 plan

Size subnets from real host counts, plus growth

1	Count real hosts per group	1 server + 1 management port + about 20 spare for containers	roughly 25 addresses
2	Choose the subnet size	/27 gives 30 usable, which is uncomfortably tight, so /24 for headroom	/24 per group
3	Allocate across the room	10.20.0.0/16 split into 10.20.G.0/24, where G is the group number 1 to 20	10.20.1.0/24 to 10.20.20.0/24
4	Reserve the special ranges at the end	10.20.250.0/24 management · 251 storage · 254 OOB	10.20.21–249 left to grow into

A rule that always pays: pick one size larger than the calculation suggests, because resizing a subnet later means reconfiguring the whole room

Designing the IPv6 plan

Not a longer IPv4, but a different way of thinking

Give every VLAN a /64

There is no need to conserve addresses; /64 is what SLAAC requires, and anything smaller breaks features

Request a /48 from the university

One /48 provides 65,536 /64 subnets — ample for the whole faculty

Tie the subnet id to the VLAN id

VLAN 101 uses 2001:db8:a:101::/64, so reading the address tells you the VLAN

Use ULA only where nothing routes out

fd00::/8 suits the OOB network, but normal networks should use real addresses without NAT

Decide how addresses are assigned

Servers static · general hosts via SLAAC · use DHCPv6 when you need a record of who got what

A security note: every interface already has a link-local address even with IPv6 unconfigured, so RA guard and DHCPv6 snooping must be on, or anyone can announce themselves as a router

The out-of-band management network

The network that must work on the day the main one does not

Why it is physically separate

Put management on the same switch as data and, when that switch fails, there is no way in to fix anything

What belongs on it

Every server's IPMI or iDRAC port · switch management ports · a console server · switchable PDUs

Control access tightly

This network reaches deepest into every machine, so it needs its own VLAN, a short access list, and logging of every session

This maps to control A.8.22 on segregation of networks: management must be clearly separated from user networks



06

Internet path and acceptance

Network work is done when it is measured, not when the lights are green

The internet path and its redundancy

Decisions to settle with the university network team

How many egress paths exist

One is a single point of failure; two must be checked not to share a single conduit

Who performs NAT, and where

You must know where internal addresses are translated, because it affects tracing logs afterwards

Where the firewall sits on the path

Whether inter-VLAN traffic crosses it affects both throughput and the rules you must request

The bandwidth actually allocated

A confirmed figure, not a shared assumption, and measured periodically

Whose DNS and NTP you use

If clocks disagree, traces and logs order events wrongly, exactly as covered in week 5

All five should be recorded in a document signed with the network team, not left as a verbal understanding

Network acceptance criteria

Eight measurable checks, not a look at the link lights

☐	Every cable passed a certification test, with results kept as a file	Cabling contractor
☐	Every port negotiated full speed and full duplex	Technical reviewer
☐	Error counters on every port read zero after 24 hours of use	Technical reviewer
☐	Every VLAN reaches what it should, and the isolated ones really are isolated	Change owner
☐	IPv4 and IPv6 addresses match the approved plan, on hosts and network gear	Change owner
☐	The OOB network is reachable while the main path is deliberately cut	Safety officer
☐	All devices agree on time to within one second via NTP	Technical reviewer
☐	Prometheus scrapes the switches and reports up = 1	Change owner

The third criterion separates good cabling from cabling that merely works — a poorly terminated cable still passes traffic while its error counter climbs

Controls that apply to the network work

In addition to the physical controls already used

A.8.20

Networks security

Networks are controlled and monitored, and network device logs are retained

A.8.21

Security of network services

Service levels and security mechanisms are agreed with the provider

A.8.22

Segregation of networks

Group, management, storage and OOB VLANs are kept apart

A.7.12

Cabling security

Cabling is protected from interception and damage; power and signal separated

A.8.16

Monitoring activities

Network traffic and anomalous behaviour are monitored

A.5.14

Information transfer

Rules for protecting information while it is in transit

References and standards

In the order worth reading them

Cabling and data-centre standards

- ANSI/TIA-942 — Telecommunications Infrastructure Standard for Data Centers
- ISO/IEC 11801-5 — Generic cabling for data centres
- ANSI/TIA-568 series — balanced twisted-pair and optical fibre cabling
- ANSI/TIA-606-C — administration standard for labelling and cabling records
- EN 50174-2 — cabling installation, including power/data separation distances

IP networking and security

- RFC 4291 — IP Version 6 Addressing Architecture
- RFC 4193 — Unique Local IPv6 Unicast Addresses (ULA)
- RFC 6434 — IPv6 Node Requirements
- RFC 7454 — BGP Operations and Security, for the external path
- ISO/IEC 27001:2022 — Annex A: A.5.14, A.7.12, A.8.16, A.8.20, A.8.21, A.8.22

This supplement in three sentences

01

Cabling is laid once and lives a decade while servers turn over every five years, so design it with more headroom than seems necessary

02

Cable dressing behind the rack is not cosmetic — it is the hot air's way out, and cables across it waste everything spent on containment

03

A good VLAN and addressing plan is one where the number tells you whose it is, with enough room that you never redesign the room

Teach this before the cabling day, then set each group to draft their own VLAN and addressing plan

A note on the artwork and figures (instructor slide)

Not for projection

Artwork in this deck

The title image, the switch-placement comparison, the rear-of-rack diagram and the VLAN table were drawn as vectors for this course and can be edited freely

The cable distance figures

Taken from TIA-568 and IEEE 802.3 as maximum channel lengths. Real installations may fall short depending on workmanship — check vendor documentation before ordering

The example IP addresses

10.20.0.0/16 and 2001:db8::/32, the documentation range per RFC 3849. Replace with your allocated ranges before use